Library walking access

Tacoma, Washington, USA

20% of residents can reach a library within a 15-minute walk (slope-aware travel time)

167,263 people are beyond that 15-minute standard

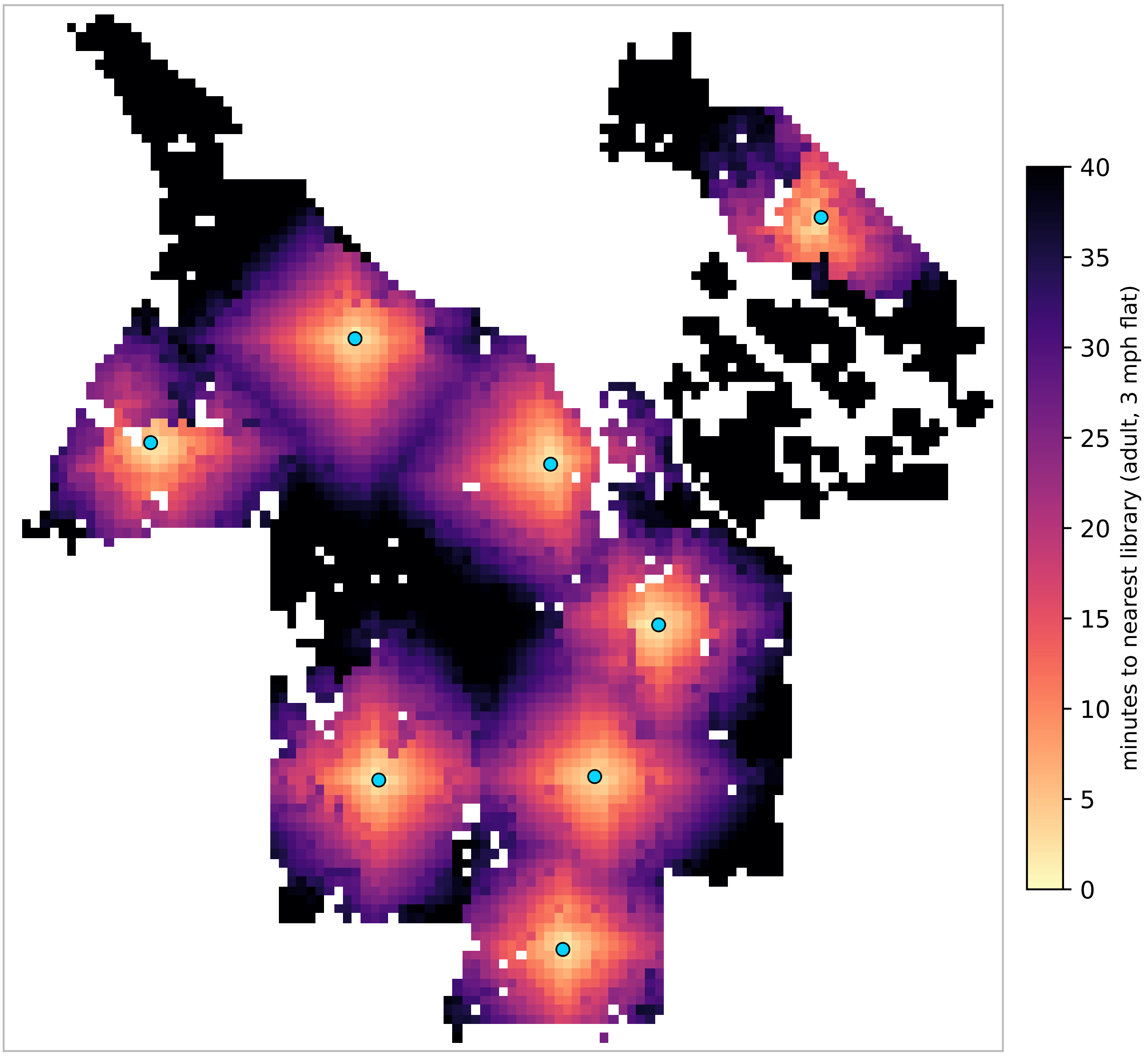
95% of residents with an ambulatory difficulty are beyond a 15-minute walk

62 min is the median walk faced by the worst-served of that group — before any slope penalty at all

Scope 8 library locations · 116 km² analysed · 210,290 residents
Terrain elevation -1–166 m · 14.2% of walk segments steeper than the 5% accessible-route grade
Standard 15-minute neighbourhood goal; 3 mph on the flat = 0.75 mi
Sources OpenStreetMap walk network · US Census ACS 5-year 2023 · USGS 3DEP elevation

Where the walk is long

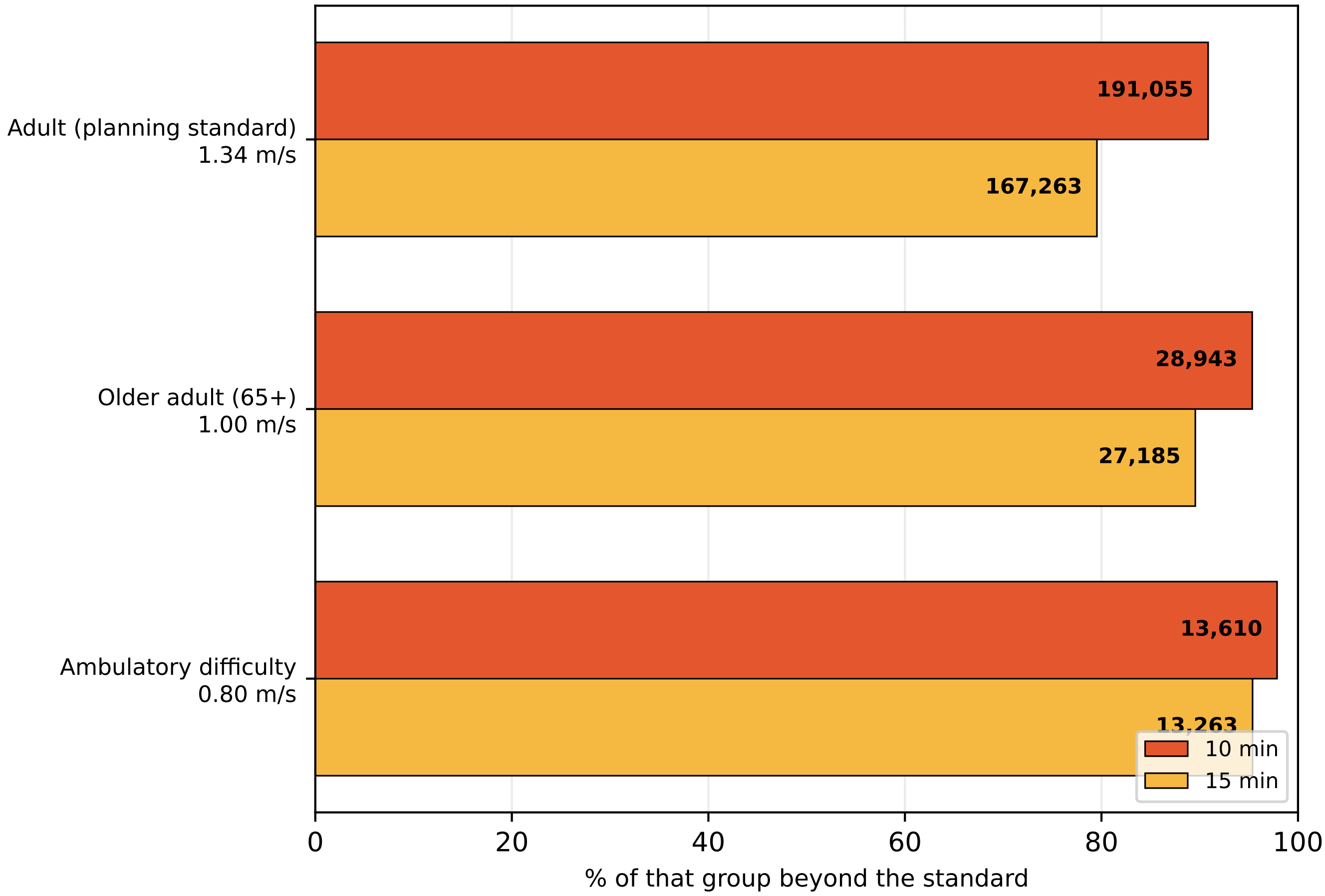
Travel time to the nearest library, on foot, accounting for slope



Median walk is 26 minutes. Blue dots are the 8 branches. Darker areas are further in time, not distance — a steep half-mile costs more than a flat one. Areas outside the analysed land mask (open water, and cells more than 250 m from any mapped pedestrian way) are blank.

A time standard is not one distance

The same policy sentence describes a different city for each kind of walker

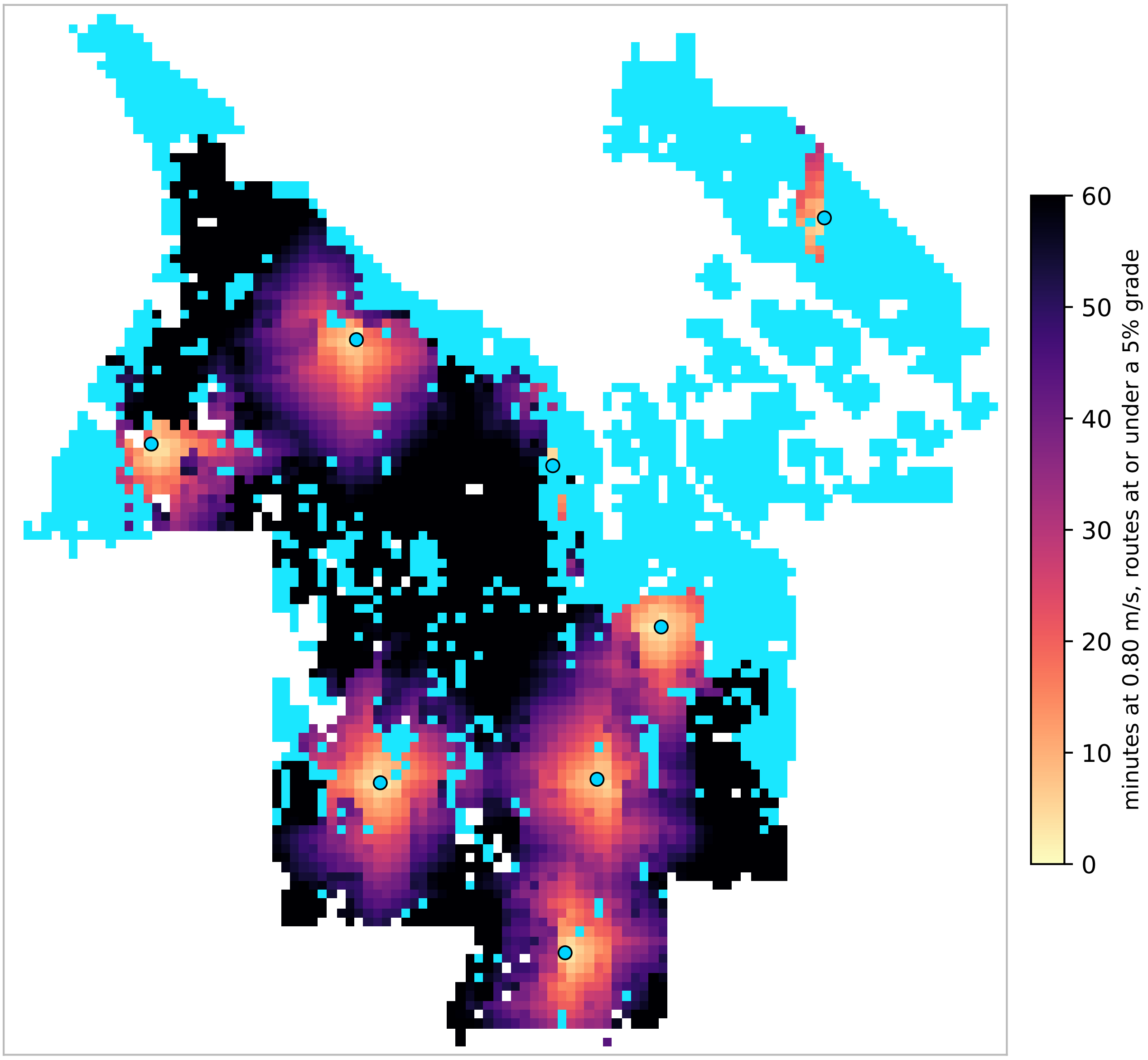


profile	group	10 min	15 min	no route
Adult (planning standard)	210,290	191,055 (91%)	167,263 (80%)	0
Older adult (65+)	30,358	28,943 (95%)	27,185 (90%)	0
Ambulatory difficulty	13,906	13,610 (98%)	13,263 (95%)	3,391

Each row is measured against its own population, not a share of the city. Walking speeds are planning defaults: 3 mph adult, 1.00 m/s for 65+, 0.80 m/s with an ambulatory difficulty.

Slope, and who it excludes

Measured against the 1:20 (5%) grade an accessible route is designed to



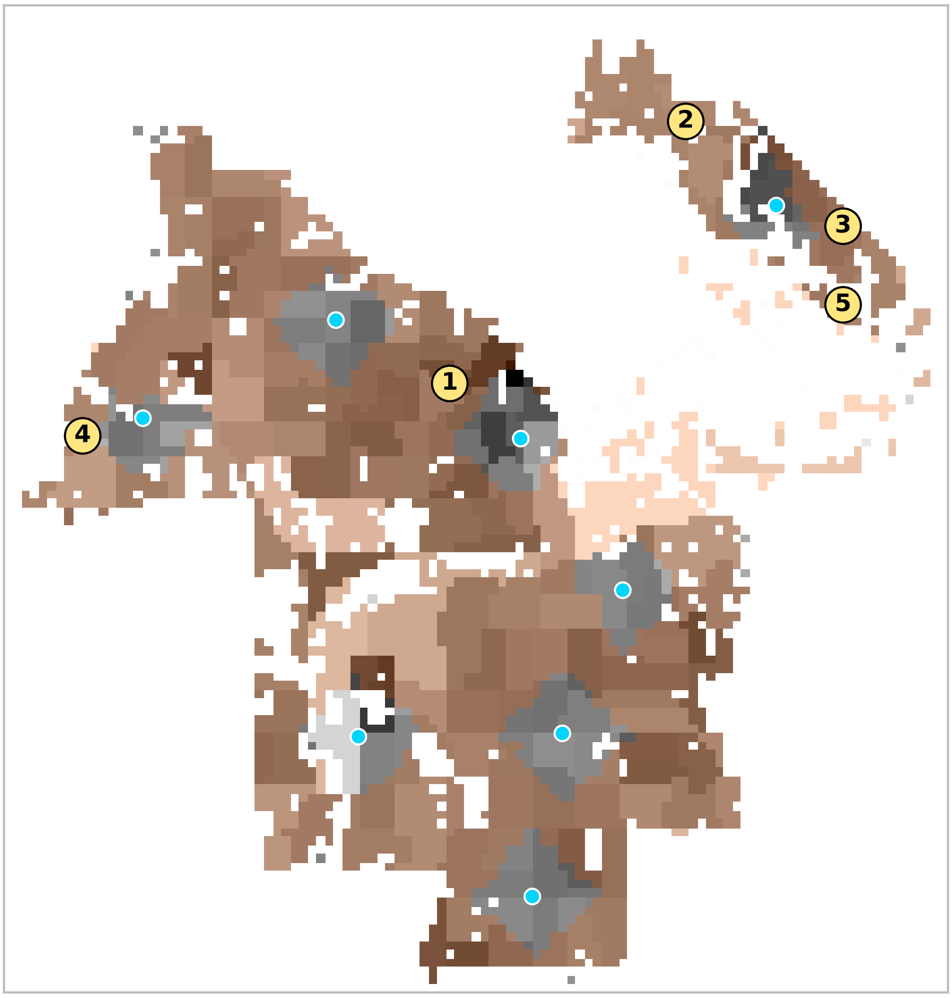
Cyan marks land with no route to any library that stays within a 5% grade: 3,391 residents with an ambulatory difficulty, 24% of that group. 14.2% of walk segments are steeper than that.

That count is sensitive to the threshold — it ranges 170–4,283 across cutoffs from 4% to 15% — so it should not be read as a precise figure. Two things underneath it are not sensitive.

These are remote places before grade is considered at all: with no slope penalty whatsoever the same cohort still faces a median 62-minute walk, and 23% over an hour. And the burden falls almost entirely on this profile — grade moves the adult figure about two points, against 5 points here.

Underserved pockets and best sites

Contiguous residential areas beyond a 15-minute walk, ranked by residents affected



#	residents	car-free HH	area	worst walk	site gains
1	147,185	5,007	63.6 km²	85 min	9,625
2	6,273	44	3.7 km²	90 min	2,913
3	5,882	24	2.2 km²	51 min	3,054
4	3,678	142	2.4 km²	47 min	1,590
5	573	3	0.4 km²	63 min	342
6	516	3	0.1 km²	34 min	0

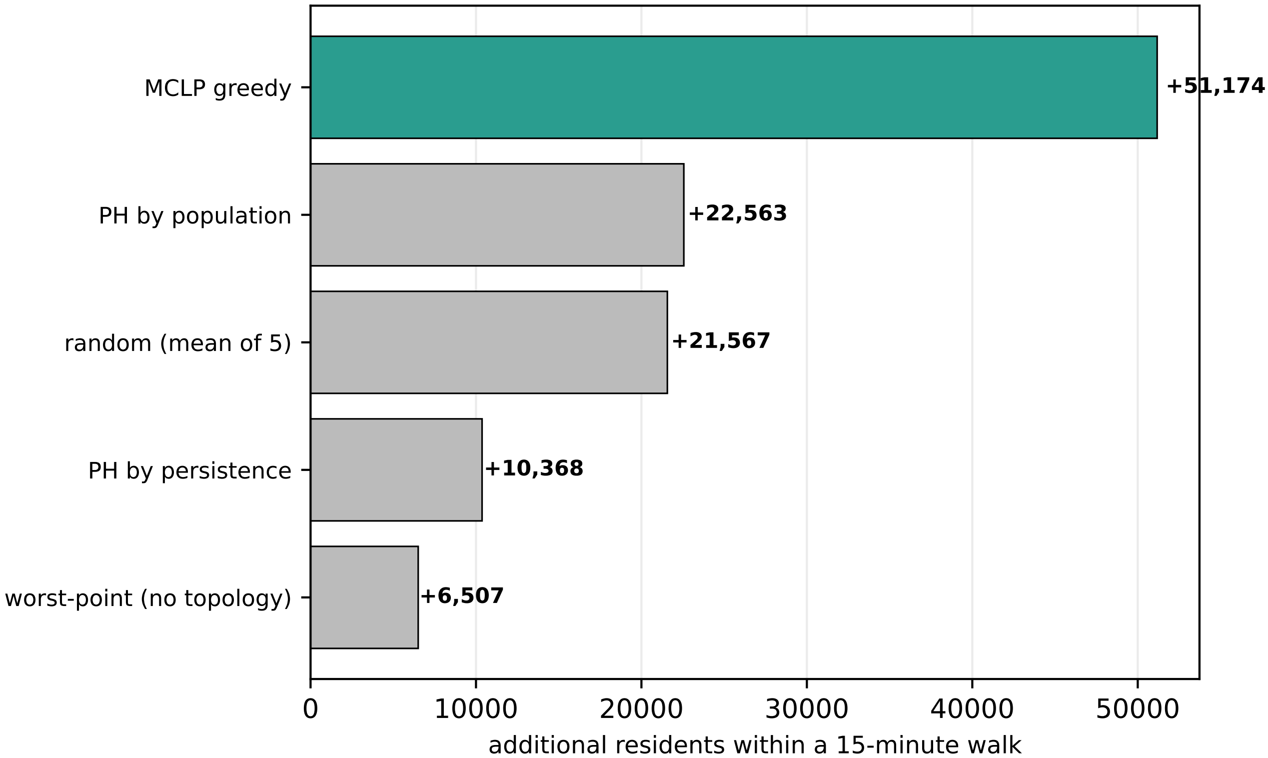
Numbered markers are the best available branch site inside each pocket — the location that brings the most residents within 15 minutes — chosen from habitable land only, so parks, port terminals and airfields are never proposed.

These are not real parcels.

A marker means “somewhere around here would help most”, resolved to a 150 m grid. It carries no assessment of zoning, ownership, lot size, acquisition cost, or whether anything is buildable on the site. Treat each one as a search area for a siting study, not a candidate address.

Where should the next 8 branches go?

Siting strategies scored on residents brought within 15 minutes



Greedy maximal-covering location (MCLP) adds 51,174 residents with 8 branches, raising coverage from 20% to 45%. It is scored here against topological and trivial alternatives; see the repository for the full comparison. Candidate sites are inhabited cells on a 450 m grid.

Method and caveats

Walk network from OpenStreetMap, retaining all connected components. Travel time uses Tobler's hiking function renormalised to each profile's flat speed; the mobility profile additionally treats segments steeper than 5% as impassable. Population is ACS 2023 block-group data rasterised to a 150 m grid; poverty, vehicle access and ambulatory difficulty are tract-level and therefore coarser.

Known limits: sidewalk quality, curb ramps, crossing delay and transit are not modelled, so the mobility figures remain optimistic even with slope. Walking speeds are literature defaults rather than locally observed. Travel time is one-way; a downhill outbound trip is uphill on return. Facility sets are branch locations only and take no account of opening hours or capacity.